



Using Tomcat DataSource

boot-jdbc-demo3 : Working Steps:

- 1) Copy **boot-jdbc-demo2** and Paste as **boot-jdbc-demo3**
- 2) Update the pom.xml

A) Exclude HikariCP

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
<exclusions>
<exclusion>
<groupId>com.zaxxer</groupId>
<artifactId>HikariCP</artifactId>
</exclusion>
</exclusions>
</dependency>
```

B) Add Dependency for Tomcat CP.

```
<dependency>
<groupId>org.apache.tomcat</groupId>
<artifactId>tomcat-jdbc</artifactId>
</dependency>
```

- 3) Update **application.properties**

A) Remove Hikari Properties

B) Add Tomcat Properties as follows

```
spring.datasource.tomcat.initialSize=5
spring.datasource.tomcat.max-active=25
```

- 4) Run the Application.

Run As => Spring Boot Application

boot-jdbc-demo3: Using Tomcat DataSource

boot-jdbc-demo3: Files required

1. MyBootApplication.java	Same As Demo2
2. CustomerDAO.java	Same As Demo2
3. CustomerDAOImpl.java	Same As Demo2
4. Customer.java	Same As Demo2
5. CustomerRowMapper.java	Same As Demo2
6. application.properties	Updated in Demo3
7. pom.xml	Updated in Demo3



6)application.properties

#Database Properties

```
spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/myjlcdb
spring.datasource.username=root
spring.datasource.password=srinivas
```

#Tomcat Properties

```
spring.datasource.tomcat.initialSize=5
spring.datasource.tomcat.max-active=25
```

#Log Properties

```
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
```

7)pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>boot-jdbc-demo3</artifactId>
<version>1.0</version>

<name>boot-jdbc-demo3</name>
<description>My Spring Boot Application</description>
<url>https://www.jlcindia.com</url>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
<exclusions>
<exclusion>
<groupId>com.zaxxer</groupId>
```



```
<artifactId>HikariCP</artifactId>
</exclusion>
</exclusions>
</dependency>

<dependency>
<groupId>org.apache.tomcat</groupId>
<artifactId>tomcat-jdbc</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

</dependencies>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



Using DBCP2 DataSource

boot-jdbc-demo4 : Working Steps:

1) Copy **boot-jdbc-demo3** and Paste as **boot-jdbc-demo4**

2) Update the pom.xml

A) Exclude HikariCP

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
    <exclusions>
        <exclusion>
            <groupId>com.zaxxer</groupId>
            <artifactId>HikariCP</artifactId>
        </exclusion>
    </exclusions>
</dependency>
```

B) Add Dependency for DBCP2

```
<dependency>
<groupId>org.apache.commons</groupId>
<artifactId>commons-dbcp2</artifactId>
</dependency>
```

3) Update **application.properties**

A) Remove Tomcat Properties

B) Add DBCP2 Properties as follows

```
spring.datasource.dbcp2.initial-size=5
spring.datasource.dbcp2.max-total=20
spring.datasource.dbcp2.pool-prepared-statements=true
```

4) Run the Application.

Run As => Spring Boot Application



boot-jdbc-demo4: Using DBCP2 DataSource

boot-jdbc-demo4: Files required

1. MyBootApplication.java	Same As Demo3
2. CustomerDAO.java	Same As Demo3
3. CustomerDAOImpl.java	Same As Demo3
4. Customer.java	Same As Demo3
5. CustomerRowMapper.java	Same As Demo3
6. application.properties	Updated in Demo4
7. pom.xml	Updated in Demo4

6)application.properties

#Database Properties

```
spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/myjlcdb
spring.datasource.username=root
spring.datasource.password=srinivas
```

#DBCP2 Properties

```
spring.datasource.dbcp2.initial-size=5
spring.datasource.dbcp2.max-total=20
spring.datasource.dbcp2.pool-prepared-statements=true
```

#Log Properties

```
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
```

7)pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>boot-jdbc-demo4</artifactId>
<version>1.0</version>
<name>boot-jdbc-demo4</name>
<description>My Spring Boot Application</description>
<url>https://www.jlcindia.com</url>
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
```



```
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
<exclusions>
<exclusion>
<groupId>com.zaxxer</groupId>
<artifactId>HikariCP</artifactId>
</exclusion>
</exclusions>
</dependency>

<dependency>
<groupId>org.apache.commons</groupId>
<artifactId>commons-dbcp2</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



Using Hikari, Tomcat and DBCP2 DataSources

boot-jdbc-demo5 : Working Steps:

1) Copy **boot-jdbc-demo4** and Paste as **boot-jdbc-demo5**

2) Update the pom.xml

A) Include HikariCP

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
</dependency>
```

B) Add Dependency for DBCP2

```
<dependency>
<groupId>org.apache.commons</groupId>
<artifactId>commons-dbcp2</artifactId>
</dependency>
```

C) Add Dependency for Tomcat CP.

```
<dependency>
<groupId>org.apache.tomcat</groupId>
<artifactId>tomcat-jdbc</artifactId>
</dependency>
```

3) Update **application.properties**

A) Add Hikari Properties as follows

```
spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5
```

B) Add Tomcat Properties as follows

```
spring.datasource.tomcat.initialSize=5
spring.datasource.tomcat.max-active=25
```

C) Add DBCP2 Properties as follows

```
spring.datasource.dbcp2.initial-size=5
spring.datasource.dbcp2.max-total=20
spring.datasource.dbcp2.pool-prepared-statements=true
```

4) Run the Application.

Run As => Spring Boot Application



boot-jdbc-demo5: Using 3 DataSourceess

boot-jdbc-demo5: Files required

1. MyBootApplication.java	Same As Demo4
2. CustomerDAO.java	Same As Demo4
3. CustomerDAOImpl.java	Same As Demo4
4. Customer.java	Same As Demo4
5. CustomerRowMapper.java	Same As Demo4
6. application.properties	Updated in Demo5
7. pom.xml	Updated in Demo5

6)application.properties

#Database Properties

```
spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/myjlcdb
spring.datasource.username=root
spring.datasource.password=srinivas
```

#Hikari Properties

```
spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5
```

#Tomcat Properties

```
spring.datasource.tomcat.initialSize=5
spring.datasource.tomcat.max-active=25
```

#DBCP2 Properties

```
spring.datasource.dbcp2.initial-size=5
spring.datasource.dbcp2.max-total=20
spring.datasource.dbcp2.pool-prepared-statements=true
```

#Log Properties

```
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
```

7)pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>boot-jdbc-demo5</artifactId>
<version>1.0</version>
```



```
<name>boot-jdbc-demo5</name>
<description> My Spring Boot Application</description>
<url>https://www.jlcindia.com</url>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat</groupId>
<artifactId>tomcat-jdbc</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

</dependencies>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



Using Tomcat and DBCP2 DataSources

boot-jdbc-demo6 : Working Steps:

- 1) Copy **boot-jdbc-demo5** and Paste as **boot-jdbc-demo6**
- 2) Update the pom.xml

A) Include HikariCP

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
<exclusions>
<exclusion>
<groupId>com.zaxxer</groupId>
<artifactId>HikariCP</artifactId>
</exclusion>
</exclusions>
</dependency>
```

B) Add Dependency for DBCP2

```
<dependency>
<groupId>org.apache.commons</groupId>
<artifactId>commons-dbcp2</artifactId>
</dependency>
```

C) Add Dependency for Tomcat CP.

```
<dependency>
<groupId>org.apache.tomcat</groupId>
<artifactId>tomcat-jdbc</artifactId>
</dependency>
```

- 3) Update **application.properties**

A) Add Tomcat Properties as follows

```
spring.datasource.tomcat.initialSize=5
spring.datasource.tomcat.max-active=25
```

B) Add DBCP2 Properties as follows

```
spring.datasource.dbcp2.initial-size=5
spring.datasource.dbcp2.max-total=20
spring.datasource.dbcp2.pool-prepared-statements=true
```

- 4) Run the Application.

Run As => Spring Boot Application



boot-jdbc-demo6: Using 2 DataSource

boot-jdbc-demo6: Files required

1. MyBootApplication.java	Same As Demo5
2. CustomerDAO.java	Same As Demo5
3. CustomerDAOImpl.java	Same As Demo5
4. Customer.java	Same As Demo5
5. CustomerRowMapper.java	Same As Demo5
6. application.properties	Updated in Demo6
7. pom.xml	Updated in Demo6

6)application.properties

#Database Properties

```
spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/myjlcdb
spring.datasource.username=root
spring.datasource.password=srinivas
```

#Tomcat Properties

```
spring.datasource.tomcat.initialSize=5
spring.datasource.tomcat.max-active=25
```

#DBCP2 Properties

```
spring.datasource.dbcp2.initial-size=5
spring.datasource.dbcp2.max-total=20
spring.datasource.dbcp2.pool-prepared-statements=true
```

#Log Properties

```
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
```

7)pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>boot-jdbc-demo6</artifactId>
<version>1.0</version>
<name>boot-jdbc-demo6</name>
<description> My Spring Boot Application</description>
<url>https://www.jlcindia.com</url>
```



```
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-jdbc</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat</groupId>
<artifactId>tomcat-jdbc</artifactId>
</dependency>

<dependency>
<groupId>org.apache.commons</groupId>
<artifactId>commons-dbcp2</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

</dependencies>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



- Q16)** What is the Default DataSource in Spring Boot 1.x?
- Q17)** What is the Default DataSource in Spring Boot 2x?
- Q18)** Can I Change the Default DataSource to other Required DataSource?
- Q19)** What to do if I want to use Tomcat dataSource in Boot 2.x?
- Q20)** What to do if I want to use DBCP2 dataSource in Boot 2.x?
- Q21)** What happens when I provide Following 3 DataSources?
HikariCP TomcatCP DBCP2
- Q22)** What happens when I provide Following 2 DataSources?
DBCP2 TomcatCP
- Q23)** What happens when I am not providing Any DataSource after excluding Hikari?